TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

Lab Report of
Introduction to Cloud Computing

A Lab Report Submitted as required for Cloud Computing

Bachelor of Science in Computer Science & Information Technology (B.SC. CSIT)

8th Semester of Tribhuvan University, Nepal

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29131/078

Submitted to:

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Symbol: 29131 Introduction to Cloud Computing

SN	Name of Practical	Date of Practical	Signature
1.	Virtualization using VMWare and Creation of Multiple Virtual Machines		
2.	Modification of Virtual Machine Resources		
3.	IP Address Identification and Network Connectivity Testing		
4.	Performing security drills and audit using Nmap		
5.	Implementation of MapReduce Program		
6.	Capturing Packets using Wireshark		
7.	Using Site24*7 monitor all necessary parameters for the MBM college.		

Q7) Using Site24*7 monitor all necessary parameters for the MBM college. You should preserve the output, for each control and monitoring step performed.

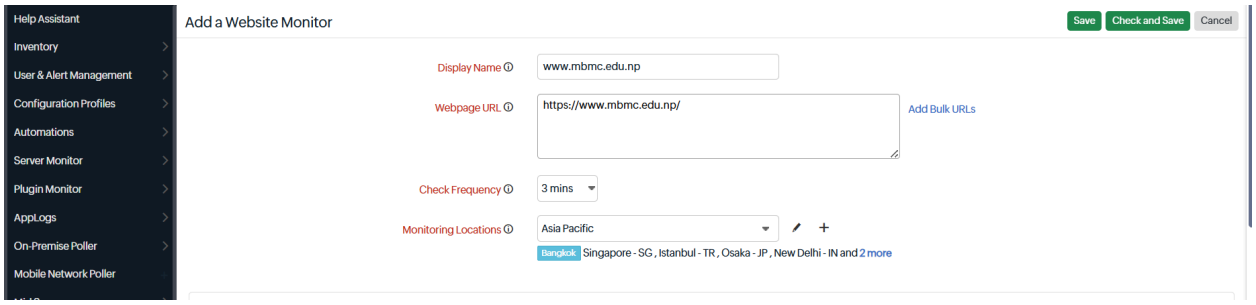
The purpose of this lab was to gain practical experience in setting up and utilizing monitoring tools to ensure the optimal performance and reliability of IT infrastructure. Specifically, we focused on using Site24x7 to monitor a Linux-based virtual machine (VM) running Ubuntu on VirtualBox. By performing this lab, we aimed to understand how to install monitoring agents, configure monitoring parameters, and analyze the collected data through the Site24x7 dashboard. This hands-on experience is crucial for IT professionals to proactively manage and troubleshoot network and system issues.

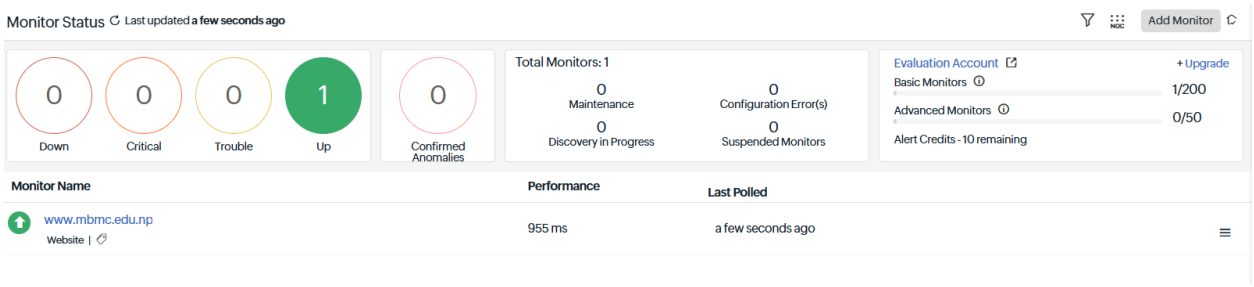
Objectives

The objectives of this lab are as follows:

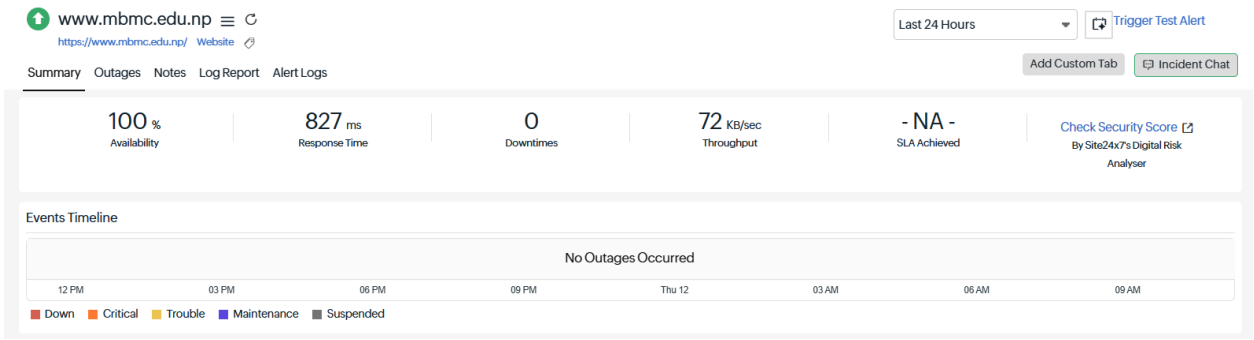
- Successfully install the Site24x7 monitoring agent on an Ubuntu VM to enable data collection.
- Configure Site24x7 to monitor essential system metrics such as CPU usage, memory usage, disk space, and network activity.
- Use the Site24x7 dashboard to review collected data, generate reports, and identify potential issues or performance bottlenecks.

Access the dashboard of site24*7 and review the reports

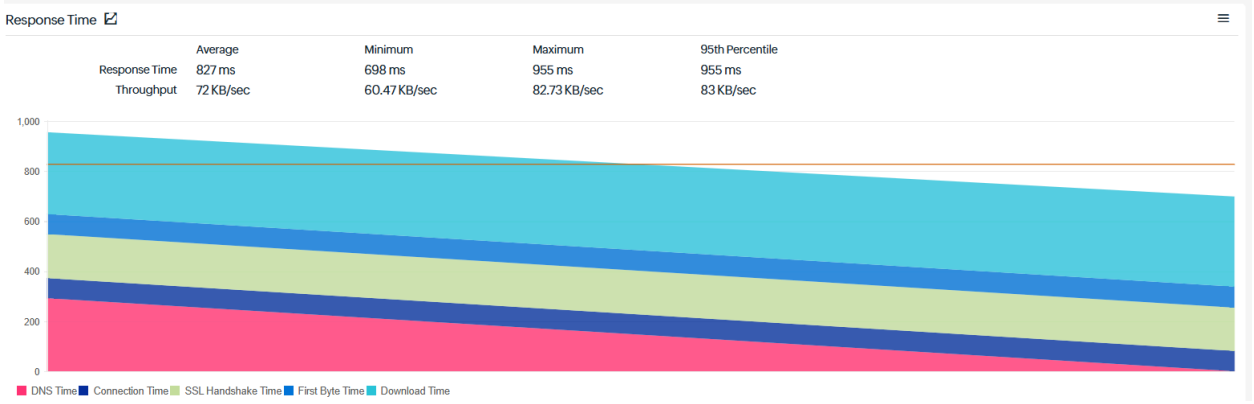




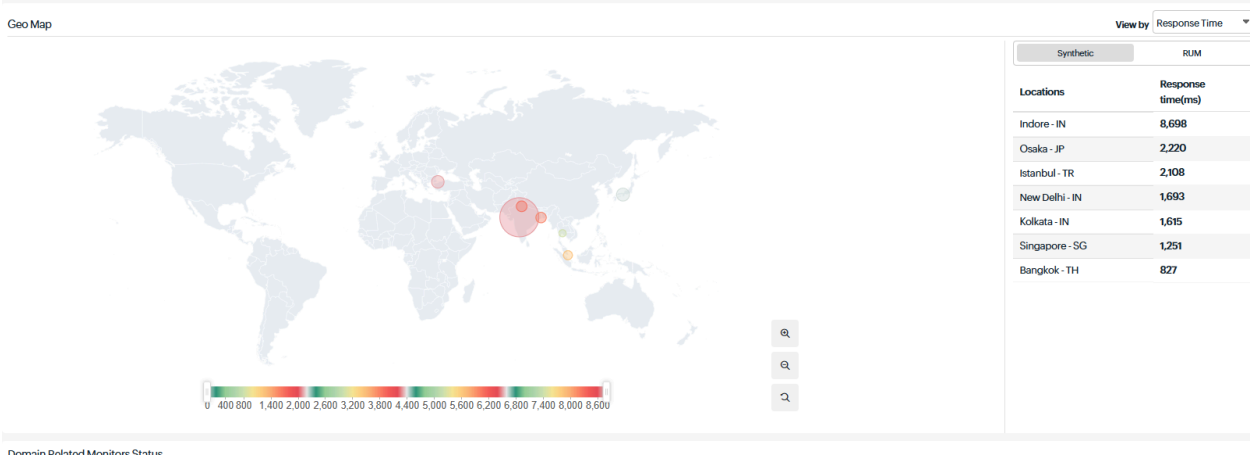
Analyzing Parameters



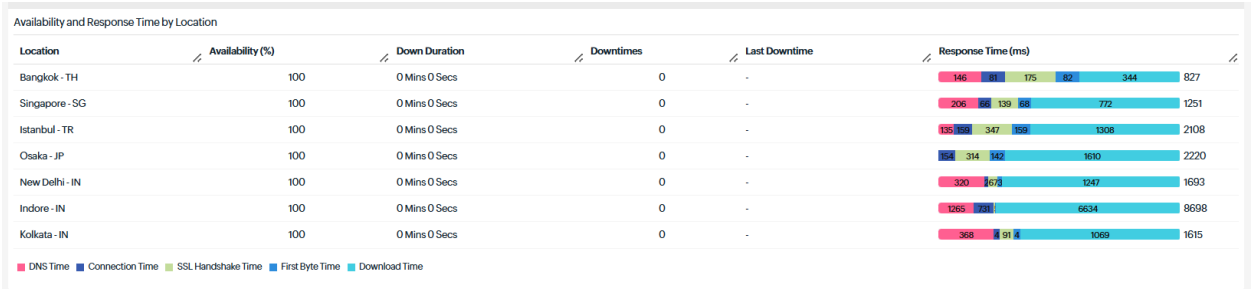
Response Time



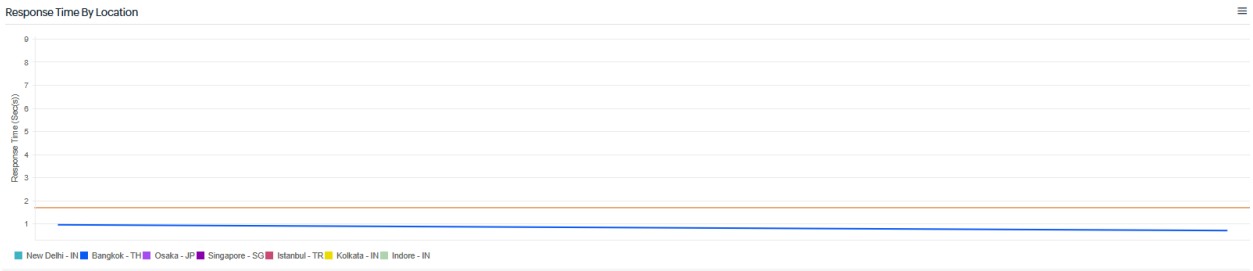
Geo Map



Availability and Response Time by Location



Response Time By Location



Global Industry Average

Global Industry Average

Response Time

Availability

Here's what you should add next to make the most of your Site24x7 account.

Server Monitoring

Monitoring your back-end servers is equally important to ensure your internet services are up and running 24x7. Install an agent and keep a check on your server performance with 60+ performance metrics.

Learn more

Add a Server Monitor

Q1) Perform virtualization using VMWare and following virtual machines: Ubuntu, Kali

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Virtualization is a fundamental concept in cloud computing that allows multiple operating systems to run on a single physical machine. This is achieved using a hypervisor such as VMware. Virtual machines share hardware resources like CPU, RAM, and storage while functioning independently. Virtualization improves resource utilization, scalability, and system management which are core characteristics of cloud computing platforms.

Key Technical Terms

Virtualization – Technology that allows multiple operating systems to run on a single hardware system.

Virtual Machine (VM) – A software-based computer system that runs an operating system just like a physical computer.

Hypervisor – Software that manages virtual machines (e.g., VMware).

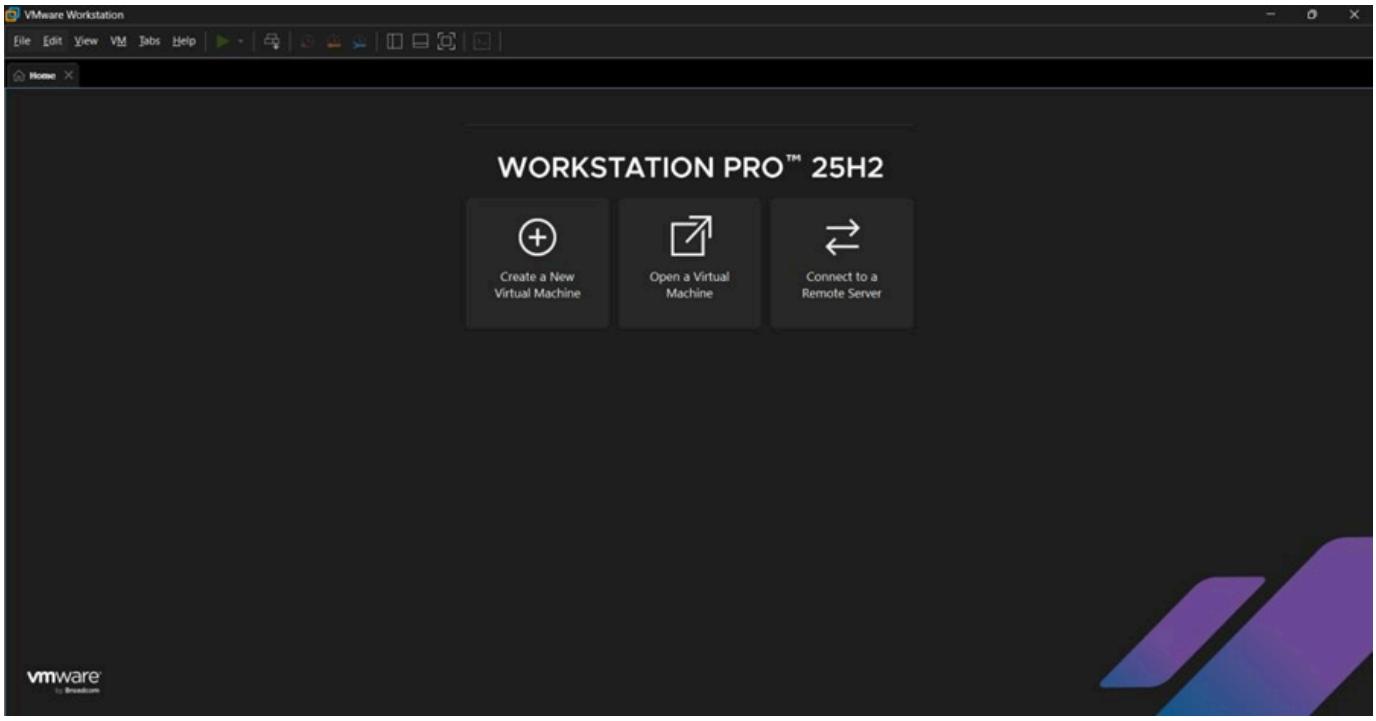
ISO Image – A disk image file used to install an operating system.

Resource Allocation – Assigning CPU, RAM, and storage to virtual machines.

Procedure

Step 1: Install VMware Workstation.

1. Download and install VMware Workstation from the official VMware website: VMware Workstation.

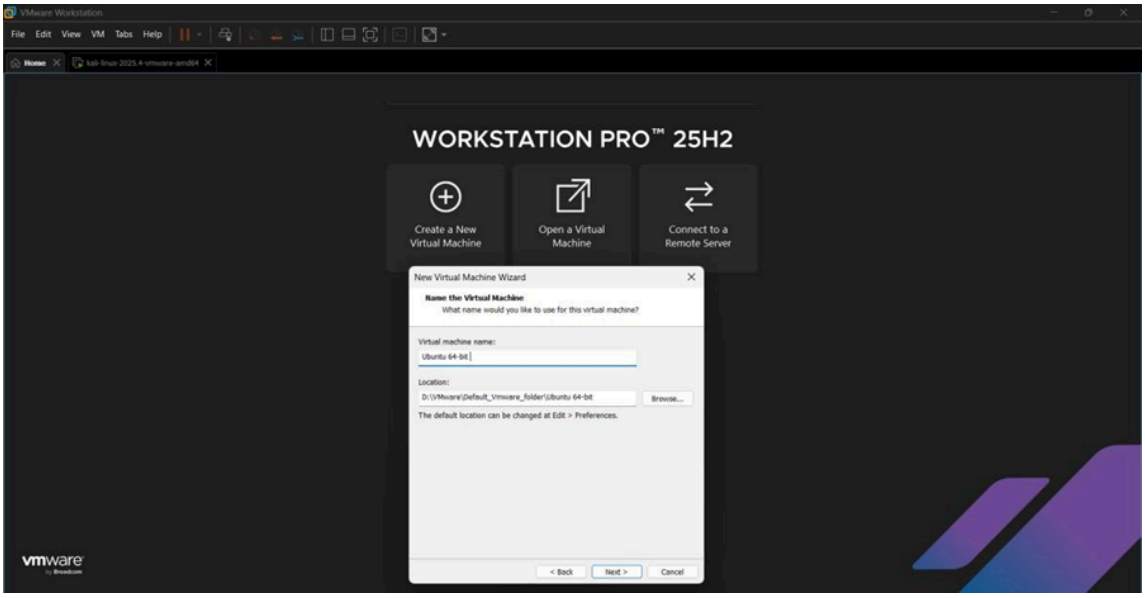


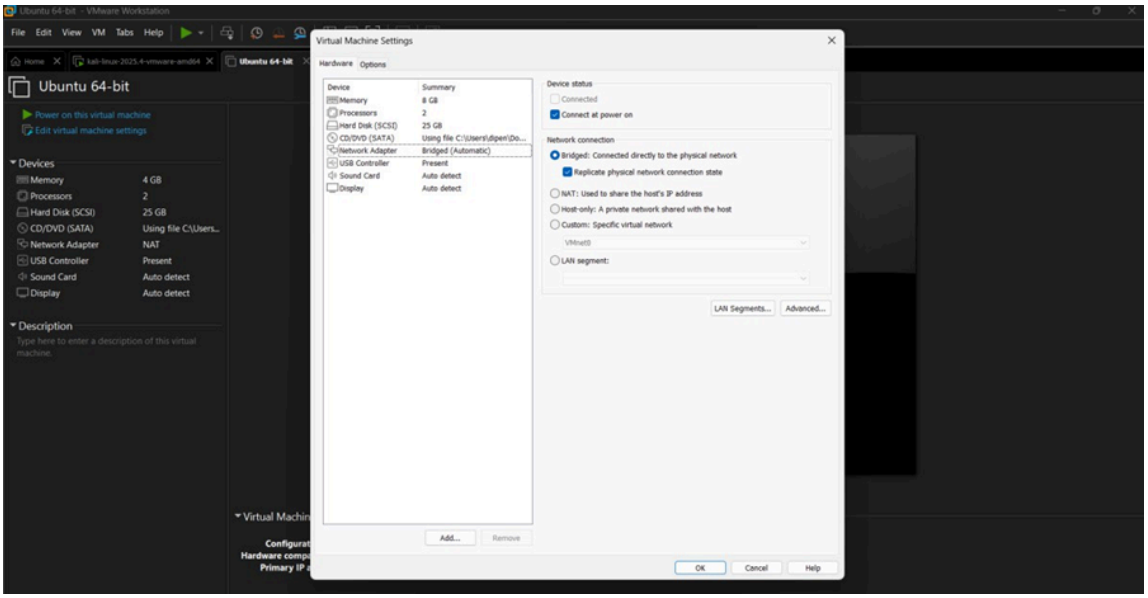
Step 2: Create a Virtual Machine(VM) for Ubuntu OS.

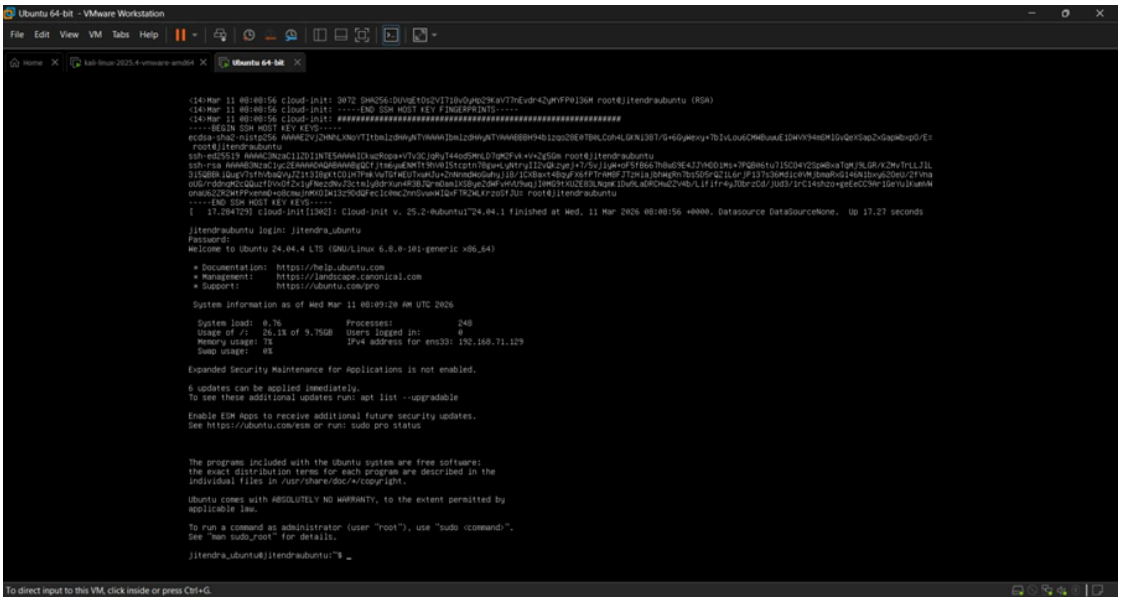
1. Launch VMware Workstation:
   - Open VMware Workstation or VMware Player.
2. Create a New VM:
   - Click "Create a New Virtual Machine".
   - Choose "Typical (recommended)" and click Next.
3. Select the OS Type:
   - Select "Installer disc image file (iso)" and browse to the Ubuntu ISO file.
   - Select Ubuntu under the operating system options.
4. Configure the VM Hardware:
   - Memory: Assign at least 2 GB RAM (or more based on your machine's capacity).
   - Processors: Assign at least 1 processor (2 cores if your system allows).
   - Storage: Assign at least 20 GB of virtual hard drive storage.
     - Select "Create a new virtual disk" and set the disk size.
5. Complete VM Creation:

- Name your VM (e.g., "Ubuntu_VM").
- Click Finish to complete the VM creation.

6. Start the VM:
   - Click Power On to boot the VM from the ISO.
   - The system will boot into Ubuntu installation. Follow the on-screen instructions to install Ubuntu.





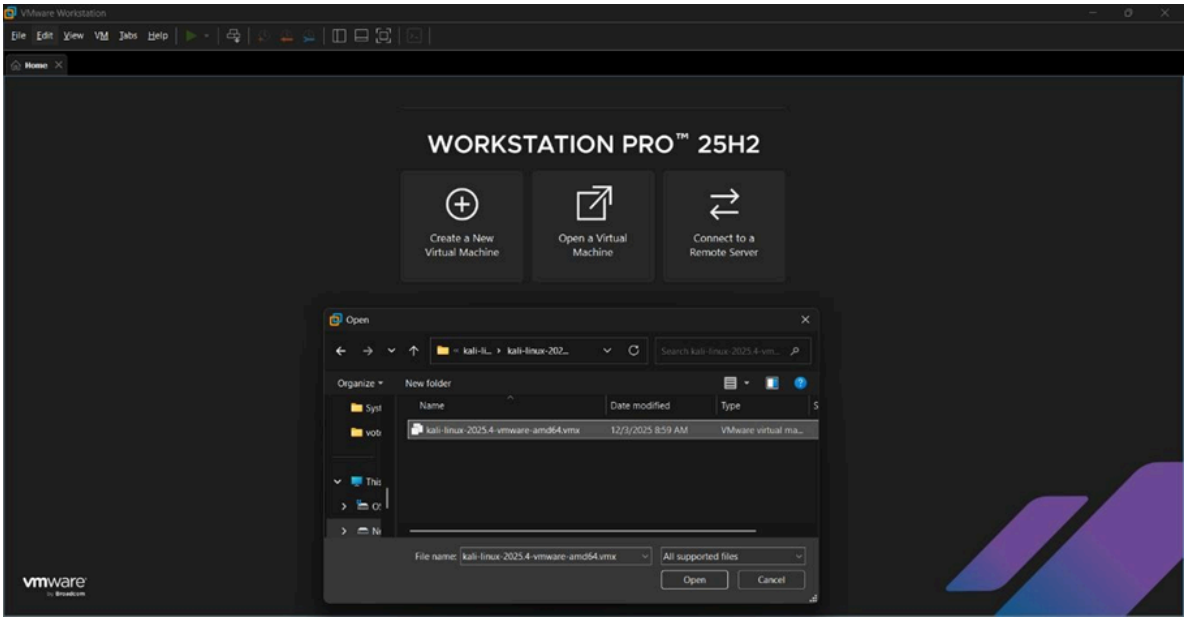


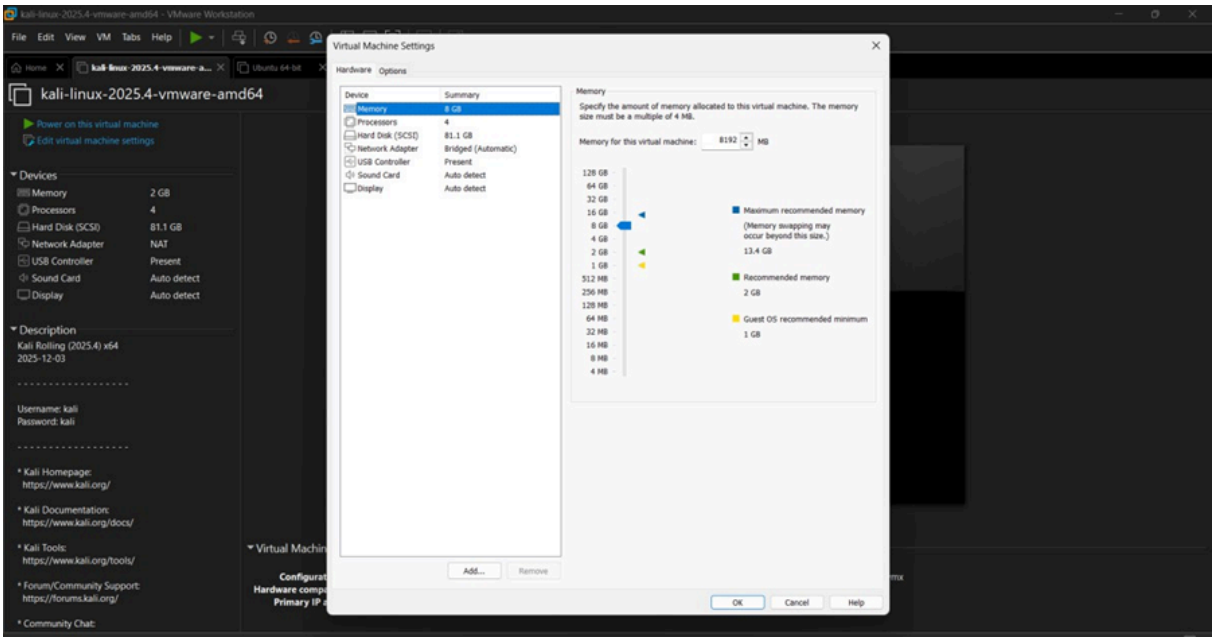
Step 3: Create a Virtual Machine for Kali Linux OS.

1. Create a New VM:

- In VMware, go to File > New Virtual Machine and choose "Typical".
- Click Next.

2. Select the OS Type:
    - Select "Installer disc image file (iso)" and browse to the Kali Linux ISO file.
    - Select Linux and then Debian 64-bit (since Kali is based on Debian).
3. Configure the VM Hardware:
    - Memory: Assign at least 2 GB RAM (or more based on your machine's capacity).
    - Processors: Assign at least 1 processor (2 cores if your system allows).
    - Storage: Assign at least 20 GB of virtual hard drive storage.
        - Select "Create a new virtual disk" and set the disk size.
4. Complete VM Creation:
    - Name your VM (e.g., "Kali_Linux_VM").
    - Click Finish to complete the VM creation.
5. Start the VM:
    - Click Power On to boot the VM from the Kali Linux ISO.
    - Follow the on-screen instructions to install Kali Linux.




kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-amd64
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 4
Hard Disk (SCSI) 81.1 GB
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Kali Rolling (2025.4) x64
2025-12-03
Username: kali
Password: kali
* Kali Homepage:
https://www.kali.org/
* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
* Forum/Community Support:
https://forums.kali.org/
* Community Chat:
Virtual Machine Settings
Hardware Options
Device Summary
Memory 8 GB
Processors 4
Hard Disk (SCSI) 81.1 GB
Network Adapter Bridged (Automatic)
USB Controller Present
Sound Card Auto detect
Display Auto detect
Add... Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine: 8192 MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.4 GB
Recommended memory
2 GB
Guest OS recommended minimum
1 GB
OK Cancel Help


kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
File System
Trash
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Q2) For the VM, change the RAM to more than 8GB and hard disk size more than 20GB, also write the observation for it.

Virtualization technology allows users to create and manage multiple virtual machines on a single physical computer. One of the major advantages of virtualization is the ability to modify hardware resources such as memory (RAM) and storage without affecting the host machine. Virtual machine managers like VMware Workstation allow administrators to dynamically adjust resources according to system requirements. Increasing RAM and disk space improves the performance and storage capacity of virtual machines

Materials and Equipment

- Computer or laptop with virtualization support
- **VMware Workstation** installed
- Previously created virtual machines:

-Ubuntu Virtual Machine

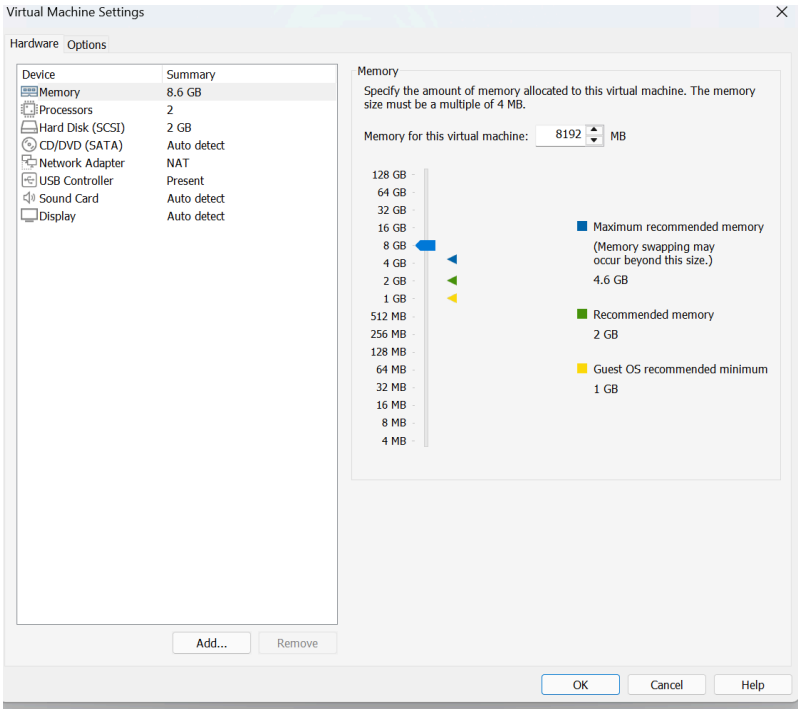
-Kali Linux Virtual Machine

- Operating system already installed in the VMs

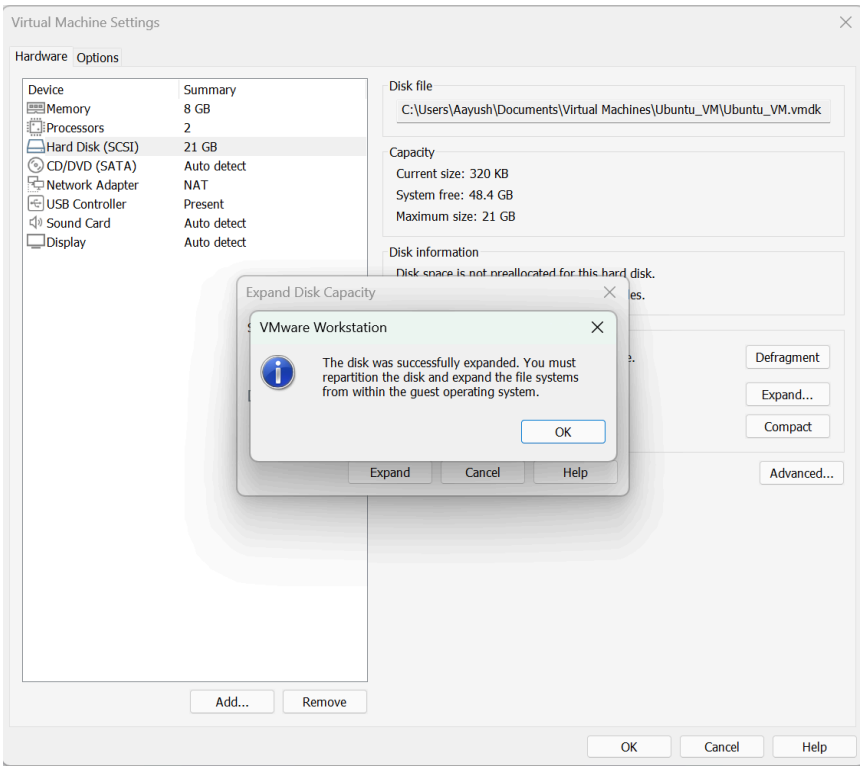
Procedure

1. Open VMware Workstation on the host computer.
2. Select the first virtual machine (Ubuntu VM) from the VMware library.
3. Click on Edit Virtual Machine Settings.
4. Select the Memory option.
5. Increase the RAM size to more than 8 GB.
6. Click on Hard Disk settings.
7. Increase the disk size to more than 20 GB.
8. Click OK to save the changes.
9. Repeat the same steps for the second virtual machine (Kali Linux VM).
10. Start both virtual machines and verify the updated hardware configuration.

Changing RAM to more than 8GB



Expanding hard disk size more than 20GB



Q3) From all VMs, show the ip address of all VMs. then ping IP address of VM.

In a virtualized environment, each virtual machine connected to a network receives a unique IP address. This IP address allows machines to communicate with each other within the network. Network testing tools such as the ping command help verify connectivity between systems. This experiment demonstrates how to identify IP addresses and test communication between two virtual machines.

Objectives

- To find the IP address of virtual machines.
- To test connectivity between two virtual machines using the ping command.

Procedure

1. Start both virtual machines in VMware.
2. In the Ubuntu VM, open the terminal.
3. Type the following command to find the IP address:

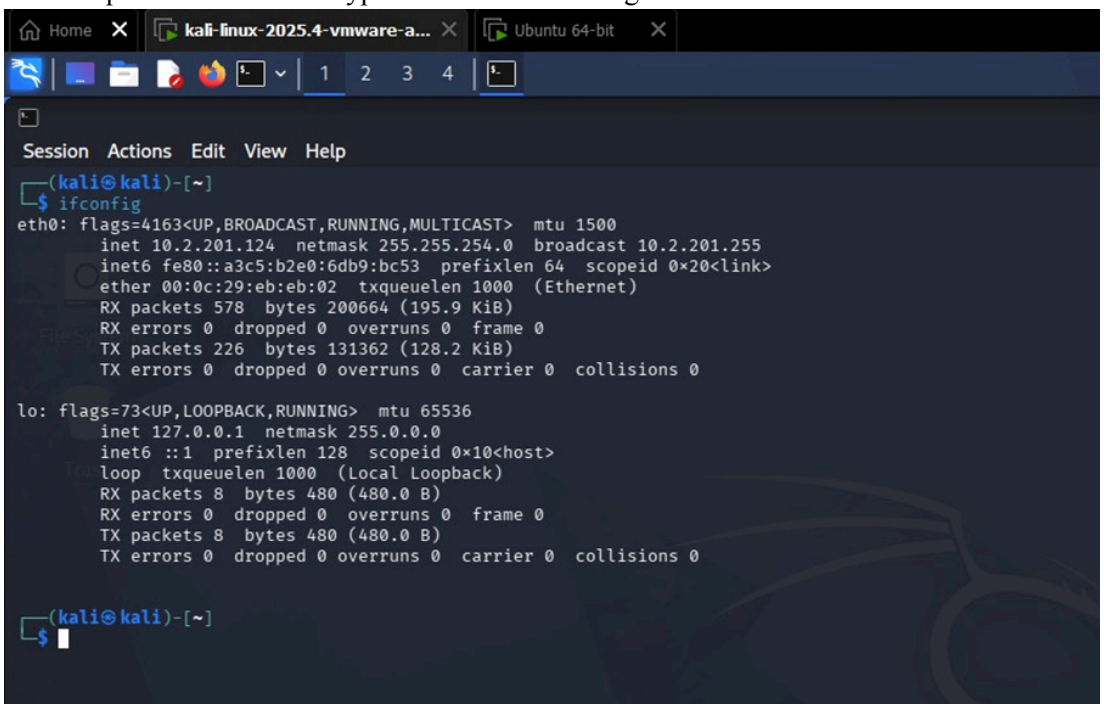
   Ifconfig or ip a

4. Note the IP address displayed.
5. Open the second VM(kali Linux or Windows).
6. Open terminal or Command Prompt.
7. Use the ping command to test connectivity:

   ping <IP address of other VM>

8. Observe whether packets are successfully transmitted and received.

1. Find out IP address of Kali Linux VM
   Opened terminal and typed command “ifconfig”.



Q5) Write a program for map reduction in any of the programming platform

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MapReduce is a programming model and processing technique for handling and generating large data sets with a distributed algorithm on a cluster. Developed by Google, it allows for the processing of massive amounts of data in a parallel, distributed manner across many computers.
Components of MapReduce:

Map Function:

Input: The input is typically a large data set divided into smaller chunks.

Processing: The Map function processes these chunks independently and in parallel. It takes a pair of input key-value pairs and produces a set of intermediate key-value pairs.

Output: The output of the Map function is an intermediate data set, which consists of key-value pairs that are grouped by key.

Shuffle and Sort:

Shuffling: This phase is responsible for transferring the intermediate key-value pairs produced by the Map function to the Reducers. It ensures that all values associated with the same key are sent to the same Reducer.

Sorting: Before the data is sent to the Reducer, it is sorted by the intermediate key. This helps in efficient grouping of values by key during the reduce phase.

Reduce Function: Input: The input to the Reduce function is the grouped intermediate data. Each group consists of a unique key and a list of values associated with that key.

Processing: The Reduce function processes each group independently. It takes the intermediate key and its associated list of values and combines them to produce a smaller set of values. Output: The output of the Reduce function is the final output of the MapReduce job. It usually consists of aggregated or summarized results.

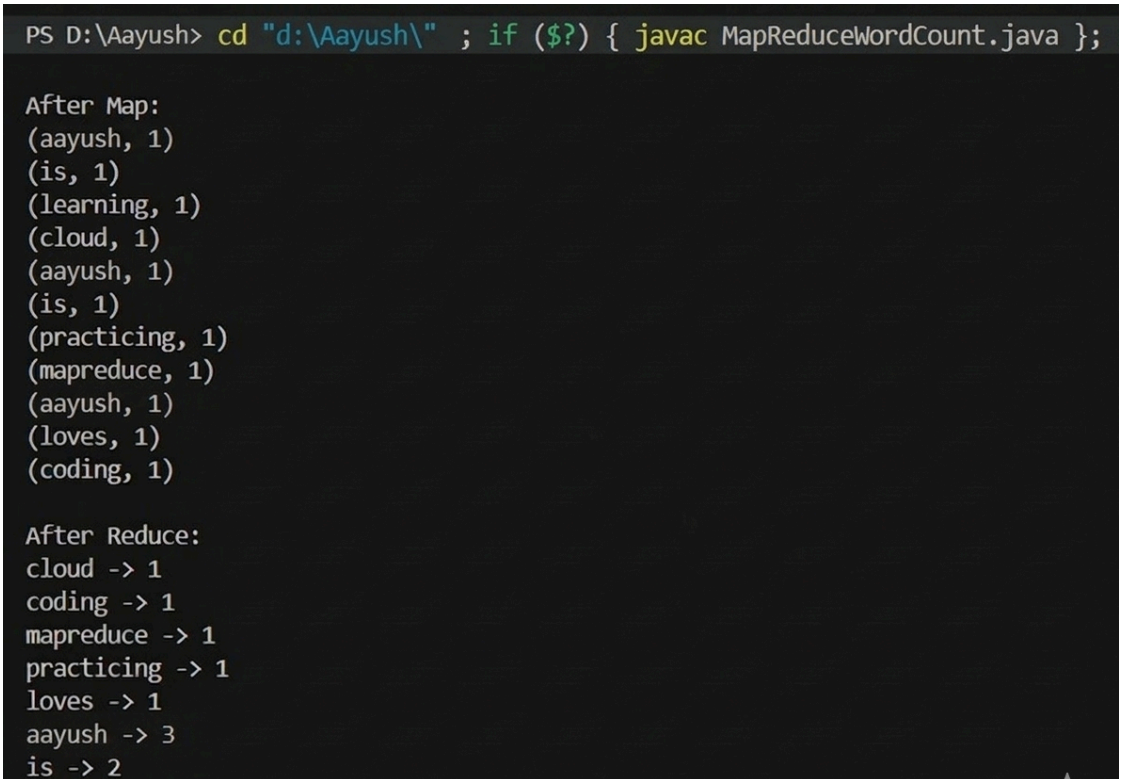
WAP of map reduce to Word Count

MapReduceWordCount.java

MapReduceWordCount.java > Language Support for Java(TM) by Red Hat > MapReduceWordCount > map(String)

```java
import java.util.*;

public class MapReduceWordCount {

    public static List<Map.Entry<String, Integer>> map(String input) {
        List<Map.Entry<String, Integer>> mappedData = new ArrayList<>();
        String[] words = input.split(regex: " ");

        for (String word : words) {
            mappedData.add(new AbstractMap.SimpleEntry<>(word, value: 1));
        }

        return mappedData;
    }

    public static Map<String, Integer> reduce(List<Map.Entry<String, Integer>> mappedData) {
        Map<String, Integer> reducedData = new HashMap<>();

        for (Map.Entry<String, Integer> entry : mappedData) {
            String key = entry.getKey();
            reducedData.put(key, reducedData.getOrDefault(key, defaultValue: 0) + 1);
        }

        return reducedData;
    }

    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        String input = "jitendra is learning cloud jitendra is practicing mapreduce jitendra loves coding";

        System.out.println(x: "Input Text:");
        System.out.println(input);
        System.out.println();

        List<Map.Entry<String, Integer>> mapped = map(input);

        System.out.println(x: "After Map:");
        for (Map.Entry<String, Integer> entry : mapped) {
            System.out.println("(" + entry.getKey() + ", " + entry.getValue() + ")");
        }
        System.out.println();

        Map<String, Integer> result = reduce(mapped);

        System.out.println(x: "After Reduce:");
        for (Map.Entry<String, Integer> entry : result.entrySet()) {
            System.out.println(entry.getKey() + " -> " + entry.getValue());
        }
    }
}
```

Output:


PS D:\Aayush> cd "d:\Aayush\" ; if ($?) { javac MapReduceWordCount.java };
After Map:
(aayush, 1)
(is, 1)
(learning, 1)
(cloud, 1)
(aayush, 1)
(is, 1)
(practicing, 1)
(mapreduce, 1)
(aayush, 1)
(loves, 1)
(coding, 1)
After Reduce:
cloud -> 1
coding -> 1
mapreduce -> 1
practicing -> 1
loves -> 1
aayush -> 3
is -> 2

Q6) Capture the packet. Capture the packet and trace the following using the appropriate methods

---

Wireshark is a widely-used network protocol analyzer that allows you to capture and interactively browse the traffic running on a computer network. It is an essential tool for network administrators, cybersecurity professionals, and developers for diagnosing network issues, analyzing traffic, and ensuring network security.

How Wireshark Works:

Capture: Wireshark captures network traffic using a packet capturing library like libpcap (on Unix-like systems) or WinPcap/Npcap (on Windows). It collects raw data packets from the network interface card (NIC).

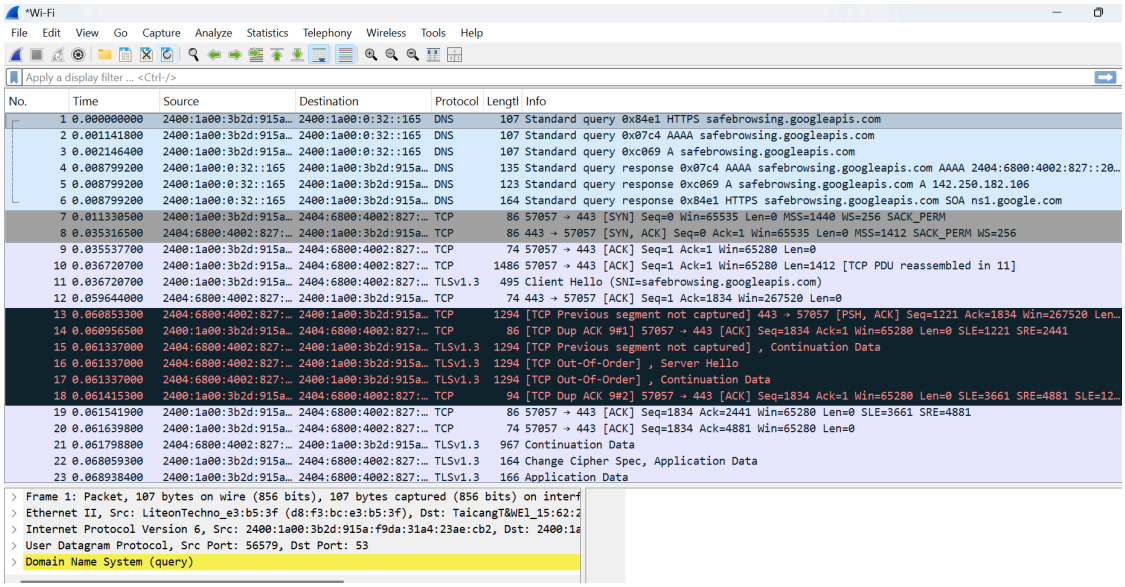
Decode: Wireshark decodes the captured packets into a human-readable format. It interprets the structure of various protocols and presents the information in a structured manner.

Display: The decoded packets are displayed in a packet list pane. Clicking on a packet reveals detailed information in the packet details pane, and the raw data is shown in the packet bytes pane.

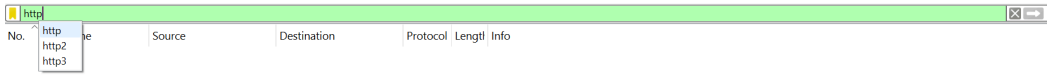
Procedure

1. Open **Wireshark.
2. Select the network interface.
3. Click Start Capturing Packets.
4. Generate traffic (open a website, ping, etc.).
5. Apply filters in the Display Filter bar.
6. Observe packets and take screenshots.

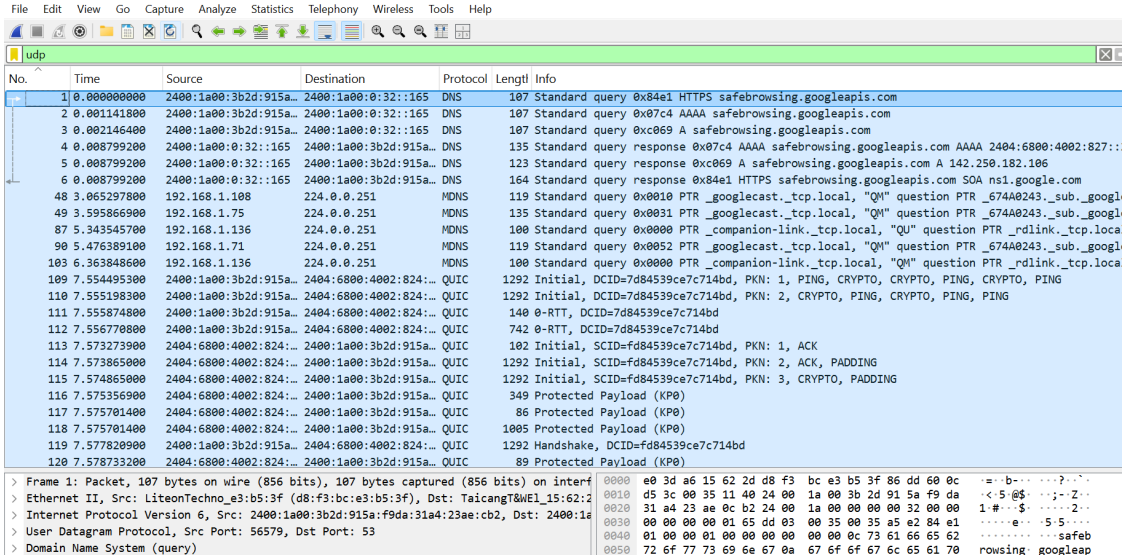
Capturing the packet



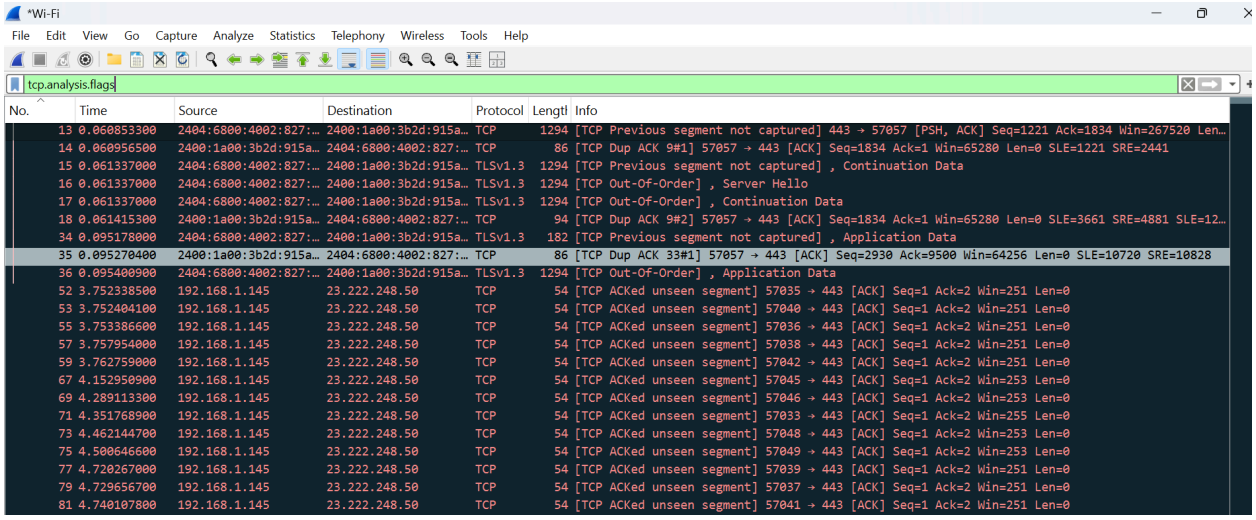
Packet that uses HTTP protocol



Packet that used UDP protocol



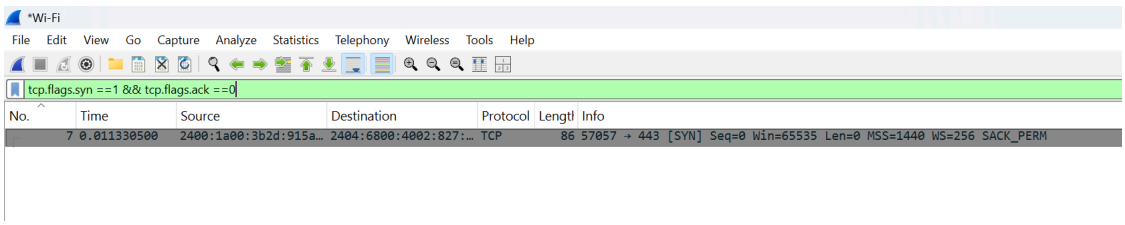
Packet that are suspicious



Packets for TCP handshaking

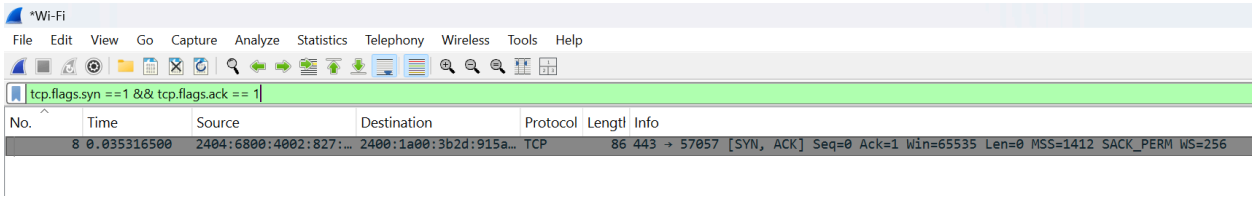
Filter for SYN

Tcp.flags.syn ==1 && tcp.flags.ack == 0



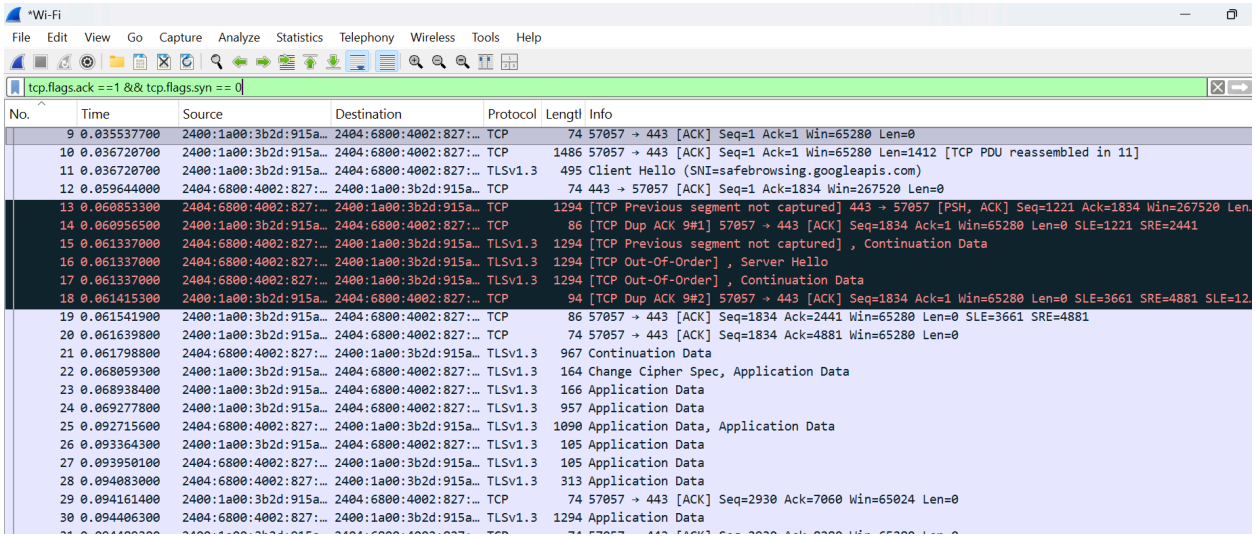
Filter for SYN-ACK

tcp.flags.syn ==1 && tcp.flags.ack == 1



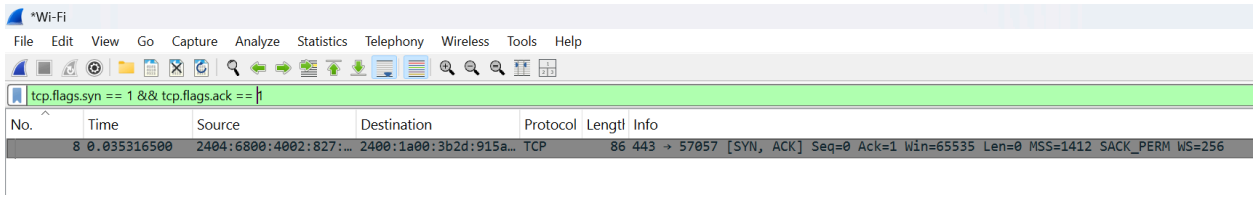
Filter for ACK

tcp.flags.ack ==1 && tcp.flags.syn == 0



Packet with syn and ack flag both

Tcp.flags.syn == 1 && tcp.flags.ack == 1



Q4) Perform security drills and audit using Nmap

This lab focuses on understanding and assessing the security posture of networks is crucial for safeguarding sensitive information and maintaining operational integrity. As part of this lab exercise, we aim to dive into the security status of two distinct networks: MBM College and Google. This investigation will utilize various specialized tools to perform comprehensive scans and evaluations, providing insights into potential vulnerabilities, configuration weaknesses, and overall network security readiness.

Objectives

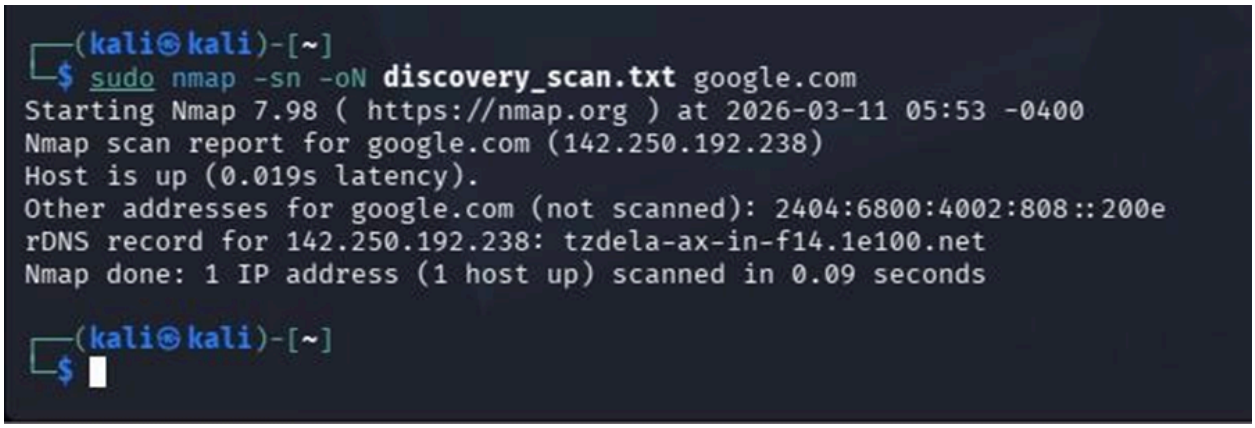
The objectives of this lab are as follows:

- Identify and map the network infrastructure of MBM College and Google using network scanning tools.
- Conduct comprehensive vulnerability assessments to detect potential security weaknesses within the networks.
- Perform web application security assessments on the web servers hosted by MBM College and Google to identify common vulnerabilities and misconfigurations.
- Document and report findings, including prioritized vulnerabilities and recommended mitigation strategies for each network.

Opened kali linux and terminal installed nmap package. Then different security drills and audits were performed. Following are the security drills and audits performed

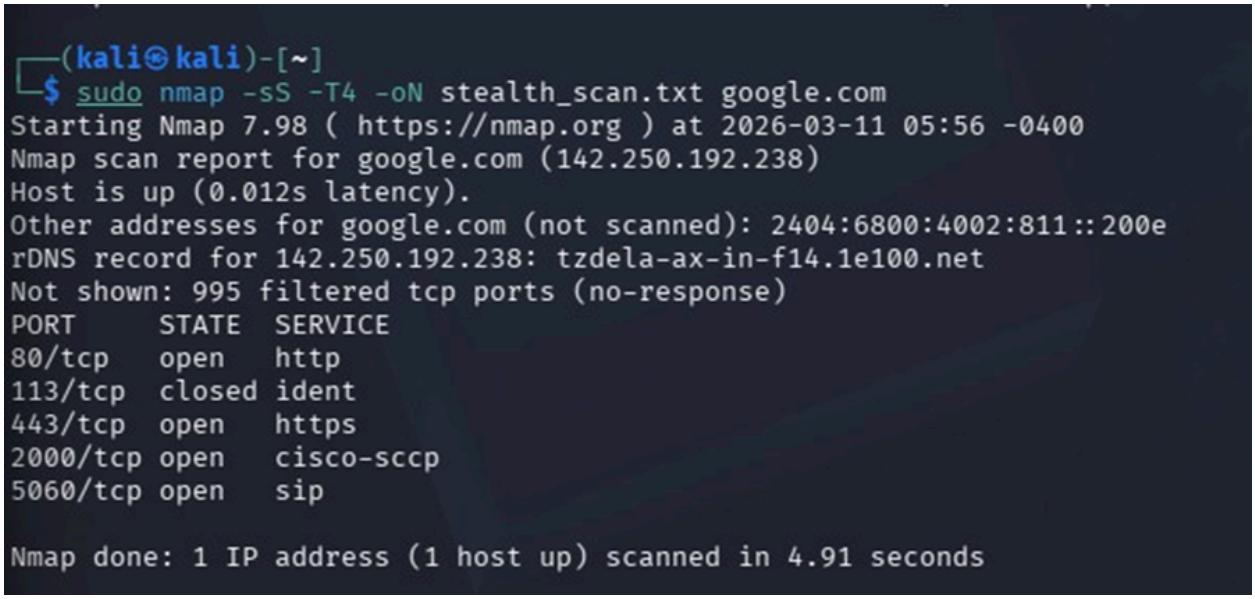
1. Hosted discovery scan (Ping sweep)

   This identifies which devices are "alive" without checking individual ports.I used command “sudo nmap –sn –oN discovery_scan.txt google.com”



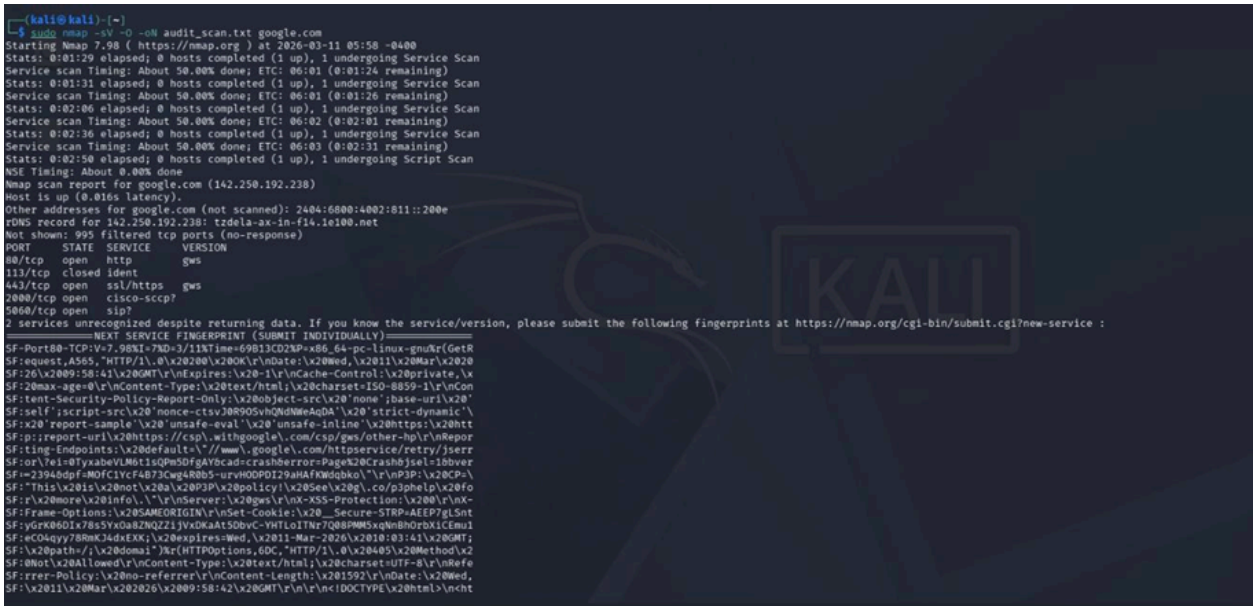
2. Stealth SYN Scan (Common Ports)

   This is the standard "silent" scan. it checks the 1,000 most common ports.I used command “sudo nmap –sS –T4 –oN stealth_scan.txt google.com”



3. Service Version & OS Detection

   This provides the "Audit" data by identifying the software versions and the operating system. I used command “sudo nmap –sY –o –oN audit_scan.txt google.com”



4. Vulnerability Audit (Scripting Engine)

   This uses the Nmap Scripting Engine (NSE) to check for known security flaws (CVEs).I used command “sudo nmap –sV –script vuln –oN vulnerability_report.txt google.com”